

Week 2 — Telemetry Analysis and Baseline

Objective

The objective of Week 2 was to analyze normal Windows telemetry and establish a baseline of expected system behavior before running any attack simulations.

This step is critical in detection engineering. Without understanding what normal looks like, it is impossible to reliably identify anomalies. Every detection threshold established in Weeks 3 and 4 was directly informed by the observations made during this baseline period.

Methodology

The Windows 10 VM was observed during normal operation with no attack activity present. The following safe actions were performed to generate representative telemetry:

```
whoami  
ipconfig  
hostname  
ping 8.8.8.8
```

A browser, Command Prompt, and PowerShell window were also opened and closed to simulate typical daily user activity.

Issues Encountered and Resolutions

Issue 1 — Sysmon EventID 1 Not Appearing in Splunk

Problem: Queries for Sysmon EventCode 1 (process creation) returned no results in Splunk during baseline analysis.

Root Cause: Sysmon process creation events were not being forwarded correctly. The channel was present in inputs.conf but the Sysmon configuration was not explicitly capturing process creation events as expected.

Resolution: Windows Security EventCode 4688 was used as a fallback for process creation detection. While EventCode 4688 provides less detail than Sysmon EventID 1 — it lacks the file hash and has limited command line capture — it provides sufficient visibility for baseline analysis and detection purposes.

Splunk query used with 4688:

```
index=windows EventCode=4688  
| table _time host "Creator Process Name" "New Process Name" "Process Command"
```

```
Line"  
| head 15
```

This query returned 314 events showing normal background process activity on the Windows endpoint.

Lesson Learned: Layered telemetry is essential. When one source fails, another should provide coverage. This is why both Sysmon and native Windows logging are deployed together in this lab environment.

Step 1 — Log Volume Analysis

The volume and distribution of log sources was assessed to understand what was being collected before any analysis began.

Query:

```
index=windows  
| stats count by sourcetype
```

Results confirmed:

Sourcetype	Count
WinEventLog:Application	~1,000
WinEventLog:Security	~1,688
WinEventLog:System	~288
WinEventLog:Windows PowerShell	~19

Total: 2,807 events across 4 active log sources.

Step 2 — Authentication Baseline

Authentication events were analyzed to understand normal login patterns before any attack activity was introduced.

Query:

```
index=windows source="WinEventLog:Security" (EventCode=4624 OR  
EventCode=4625)  
| stats count by EventCode  
| sort - count
```

Observations:

- EventCode 4624 (successful logon) appeared regularly from local interactive sessions and Windows service logons

- EventCode 4625 (failed logon) did not appear at all during the baseline period
- Logon Type 2 (interactive) and Logon Type 5 (service) were the most common types observed

Logon type analysis:

```
index=windows source="WinEventLog:Security" EventCode=4624
| stats count by Logon_Type
| sort - count
```

Detection implication: Any EventCode 4625 event is immediately anomalous against this baseline. A burst of five or more failures from a single source IP is a reliable brute force indicator.

Step 3 — Process Creation Baseline

Query used:

```
index=windows EventCode=4688
| table _time host "Creator Process Name" "New Process Name" "Process Command Line"
| head 15
```

This query returned 314 events confirming normal background process activity on DESKTOP-6C9QB2R.

Commonly observed processes under normal conditions:

Process	Frequency	Description
svchost.exe	Very High	Windows service host
taskhostw.exe	Medium	Task host process
RuntimeBroker.exe	Medium	App permissions broker
explorer.exe	Low	Windows shell
cmd.exe	Low	When opened manually by user
powershell.exe	Low	When opened manually by user
conhost.exe	Low	Console host for terminal windows

Detection implication: cmd.exe and powershell.exe are normal when opened directly by a user but suspicious when spawned by unexpected parent processes such as a downloaded executable. Parent-child process relationships carry more detection value than individual process names alone.

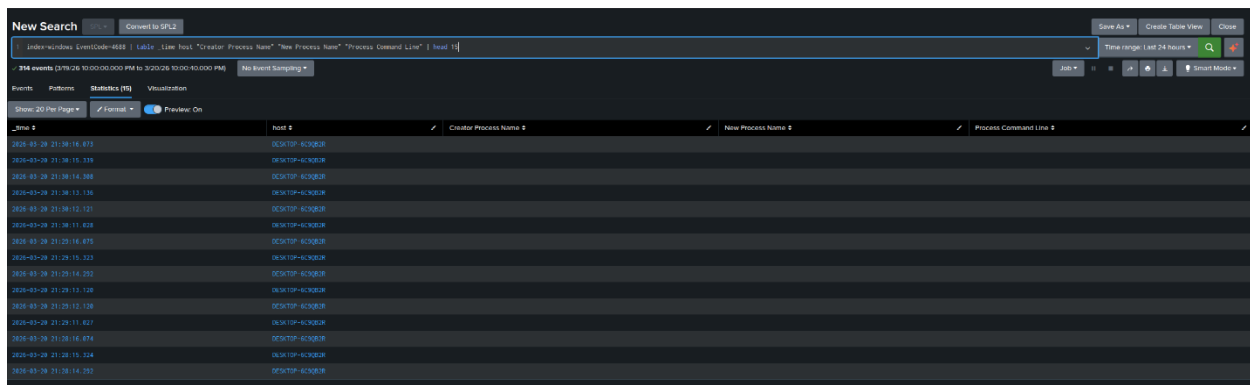


Fig 1: EventCode4688

Step 4 — Network Connection Baseline

Query:

```
index=windows EventCode=3
| stats count by Image DestinationIp DestinationPort
| sort - count
```

Commonly observed connections under normal conditions:

Process	Port	Description
svchost.exe	443	Windows Update and telemetry
chrome.exe	443	Browser HTTPS traffic
svchost.exe	53	DNS resolution
MsMpEng.exe	443	Windows Defender cloud lookups

Detection implication: All legitimate outbound traffic observed used standard ports 80, 443, and 53. Any outbound connection on a non-standard port such as 4444 has no legitimate explanation in this environment and is a reliable indicator of malicious activity.

Step 5 — Baseline Dashboard

A simple baseline dashboard was created in Splunk to visualize normal system behavior. The following queries were used to populate its panels.

Login activity over time:

```
index=windows source="WinEventLog:Security" EventCode=4624
| timechart count
```

Process creation activity:

```
index=windows EventCode=4688
| timechart count
```

Network connections over time:

```
index=windows EventCode=3
| timechart count
```

Top network-active processes:

```
index=windows EventCode=3
| stats count by Image
| sort - count
```

Parent-child process relationships:

```
index=windows EventCode=4688
| table _time host "Creator Process Name" "New Process Name" "Process Command Line"
| sort _time
```

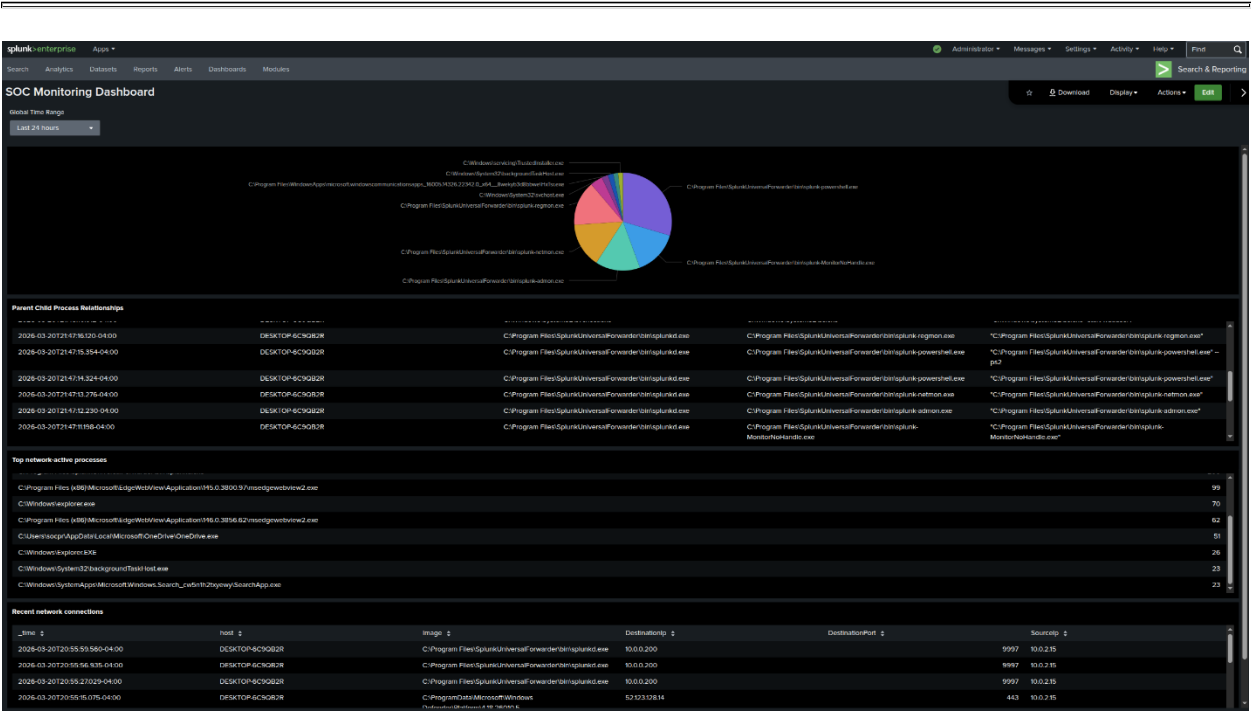


Fig 2: Initial Dashboard

Baseline Summary

Event ID	Normal Frequency	Detection Threshold
4624	Low — service logons	Logon Type 3 or 10 is anomalous
4625	None	Any occurrence is anomalous
4688	Medium — system processes	Unusual parent process is anomalous

Event ID	Normal Frequency	Detection Threshold
4698	None	Any occurrence is anomalous
4720	None	Any occurrence is anomalous
Sysmon 3	Medium — standard ports	Non-standard port is anomalous
Sysmon 13	Very High — system noise	Run key modification by non-system process is anomalous

Key Lessons Learned

1. Baseline before attacking. Without a baseline, there is no reference point for normal behavior. Detection thresholds were set based on observed baseline frequencies rather than arbitrary numbers.

2. Not all telemetry is equal. Sysmon EventID 13 generates far more noise than any other source in this environment. High-volume events require aggressive filtering to be useful for detection.

3. Absence of events is also a baseline. EventCode 4625, 4698, and 4720 did not appear at all during normal operation. Any occurrence of these events warrants investigation regardless of volume.

4. Telemetry gaps must be identified early. The absence of Sysmon EventID 1 was discovered during baseline analysis rather than during an active attack investigation, allowing a fallback detection method to be prepared in advance.

Week 2 Outcome

By the end of Week 2 the following was completed:

- Normal authentication patterns documented
- Normal process creation activity documented
- Normal network connection patterns documented
- Registry modification noise levels assessed and understood
- Baseline detection thresholds established for all key event types
- Splunk baseline dashboard created
- Sysmon EventID 1 telemetry gap identified and documented with a fallback method prepared